AMSS 2026 — Lecture 1: Intro + the AI-mediated SDLC

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Welcome

- AMSS 2026 — Analiza și Modelarea Sistemelor Software
- *Ediția AI-mediated*
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- Semester: Fall 2026

Today's Agenda

1. Frame: the architect-critic loop + 5 course commitments
2. Live demo: an AI-driven design loop on a tiny scenario
3. What AI changes at each SDLC stage
4. Tooling preview
5. Course logistics

AI Is Changing the SDLC

- Requirements, models, code, tests — all increasingly AI-generated.
- The skill is no longer *producing* artifacts. It's *directing* AI to produce them and *judging* what comes back.
- This course teaches both halves.

The Architect-Critic Loop

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- **Architect** — drive AI through the SDLC. Prompt, redirect, choose what to keep.
- **Critic** — read AI's output. Spot fabrication. Spot wrong multiplicity. Spot decorative pattern application.

Your Role: Both Halves

You are simultaneously:

- **Architect / director (B):** drive the work
- **Critic / reviewer (A):** read the work

The architect-half makes you the owner of the design. The critic-half makes you the owner of the quality.

Five Course Commitments (1/5)

AI is the default tool, not an exception.

Every artifact in this course — requirements, models, code, tests — is normally generated *with* AI. Your value-add is direction, judgment, and correction.

Five Course Commitments (2/5)

The trail is graded, not the running system.

What gets graded: directed-design narrative + final reviewed artifact set + oral defense. What does *not* get graded: running code, raw test pass-rates, hand-drawn diagrams.

Five Course Commitments (3/5)

Implementation is a forcing function, not a deliverable.

TDD-with-AI is *required* for the project — but its output is *never* a grading line item.

The reflection on what TDD revealed about your spec is what gets read.

Five Course Commitments (4/5)

Defensible literacy floor — F1 + F3 + F4.

In the oral defense, *unaided*, you must demonstrate:

- **F1** — Read & critique any AI-generated UML diagram on the spot.
- **F3** — Articulate why you directed AI a certain way and what you accepted or rejected.
- **F4** — Defend traceability across your project: requirement → use case → class → state/sequence → test.

Five Course Commitments (5/5)

Tooling parity.

Every student uses the same canonical agentic tooling: one editor extension, one model endpoint, one config file.

You may *also* use Claude Code, Copilot, Cursor on personal accounts — but graded artifacts must reproduce on the canonical setup.

Watch For Both Moves

In the demo coming up:

- When I'm typing prompts $\rightarrow$ that's the **architect** move.
- When I'm reading AI's output and pointing at problems $\rightarrow$ that's the **critic** move.

You're going to do both halves yourself, starting in Lab 1.

Demo: Library Kiosk

Live AI design loop. Watch the architect move and the critic move.

Prompt to AI: *"Generate a UML class diagram for a small library kiosk: users borrow and return books; staff register returns; books can be reserved while on loan."*

What AI Changes at Each SDLC Stage

The next 7 slides walk the SDLC stage by stage.

For each: **what AI does well** vs. **where AI fails**.

This is also the rest of your semester — each stage previews a future week.

Requirements Gathering (W2 preview)

AI does well:

- Drafts long lists of plausible-looking requirements fast.
- Surfaces stakeholder roles and use cases you might not have considered.

AI fails at:

- Fabricating non-functional requirements without grounding.
- Vague NFRs ("the system shall be performant").
- Over-specifying — generating 200 requirements where 20 would do.

Test/Spec Design (W3 preview)

AI does well:

- Generates test scaffolds from a clear specification.
- Surfaces edge cases the spec doesn't mention.

AI fails at:

- Producing tests for vague specs — they pass for the wrong reasons.
- Over-fitting tests to a particular implementation.
  If AI can't produce a passing test from your spec, your spec is too vague.

Structural Modeling (W4-5 preview)

AI does well:

- Drafts a class diagram from a one-page spec in seconds.
- Suggests reasonable class names and relationships.

AI fails at:

- Wrong multiplicities (you saw one in the demo).
- Fabricated classes with no behavior.
- Conflating abstract concepts with concrete instances.

Behavioral Modeling (W6-7 preview)

AI does well:

- Generates sequence diagrams for happy-path scenarios.
- Sketches state machines for object lifecycles.

AI fails at:

- Fabricating messages between objects that don't exist.
- Missing guards on state transitions.
- Orphan states / unreachable transitions.

Patterns (W8-9 preview)

AI does well:

- Recognizes textbook pattern situations and applies them.
- Knows the GoF vocabulary.

AI fails at:

- Decorating code with patterns that solve nothing.
- Overusing patterns where simple code would do.
- Calling something “Visitor” when it’s just an `if/else` on a type tag.

Quality, Traceability, Evaluation (W10-11 preview)

AI does well:

- Cross-references artifacts on demand.
- Spots structural gaps if asked the right question.

AI fails at:

- Maintaining traceability across iterations.
- Self-evaluating its own output.
- Catching consistency issues without being prompted.

Coding (Out of Scope Here)

Implementation is a *parallel course's* domain — not ours.

We use it as our **forcing function**: TDD-with-AI is required for the project, but its output is never graded.

What gets read is your *reflection on what the loop revealed*.

Back to F1 + F3 + F4

Every stage above gets evaluated through the same three abilities:

- **F1** — read & critique what AI produced.
- **F3** — articulate why you directed AI the way you did.
- **F4** — defend the trace from one stage to the next.

This is the literacy floor. The oral defense tests it directly.

What You Just Saw: The Course Tooling

- **Editor:** VS Code with the Continue.dev extension.
- **Endpoint:** the canonical course model endpoint (configured in `tooling/.continue/config.yaml`).
- **Mode:** agentic chat — file-and-repo aware, not browser ChatGPT.

This is what every student in the cohort runs.

Where We Install: Lab 1

- Lab 1 (Week 2): hands-on tooling onboarding. Don't try alone before then.
- Setup guide: `tooling/SETUP.md` in the course repo.
- Estimated time: ~30 min on a working laptop.

Tooling Parity (BYO Allowed, On Top)

You may also use, on your personal accounts:

- Claude Code, GitHub Copilot, Cursor — anything you have access to.

But: every graded artifact must reproduce on the canonical setup.

This is what lets the examiner reproduce your work during the oral defense.

Schedule

- **14 weeks** of lectures, one per week (100 min each).
- **7 labs**, biweekly (Weeks 2, 4, 6, 8, 10, 12, 14 — 100 min each).
- Course landing page: `traiansf.github.io/class/amss2026/`.

Final Grade

Component	Points
Project	8
Attendance	1
Din oficiu	1
Total	**10**

The Project

- **Teams of 3-5 students.**
- Domain announced by **31 October 2026**.
- Multiple teams may share a domain — the directed-design trails differ.
- Each student owns a **slice** of their team's system.
- December checkpoint (Lab 5): 1 of the 8 project points.
- Final defense (W14 + Lab 7): the rest.

Oral Defense — F1 + F3 + F4

Cold defense, 3 of 8 project points. *Unaided*, you must:

- **F1** — Read & critique any AI-generated UML diagram from any team's repo.
- **F3** — Articulate rationale for your own design decisions.
- **F4** — Defend traceability across your project's slices.

This is the integrity check.

Resit Exam (Restanță)

For students who fail the regular evaluation path:

- Single 90-min written paper.
- Format: critique an AI-generated artifact set + short-answer rationale + traceability walk.
- **No AI in the room.**
- Tests the same competence (F1+F3+F4) as the oral defense.

Academic Integrity

- AI use is **expected** for almost every artifact in this course.
- The *directed-design narrative* is your record of what you did with AI.
- The *cold defense* is the integrity check — examiners verify you understand your own work.
- One-page statement attached to the syllabus (circulated before W2).

Communication

- *Teams link / mailing list — to be circulated before W2.*
- Office hours: by appointment (email).
- Course repo: GitHub (link circulated with the team-formation announcement).

That's It For Today

- Next week: requirements with AI.
- Lab 1 (Week 2 lab slot): tooling onboarding + first AI-driven requirements drill.

Questions?